

Question 4 (10%)

Consider the function $f = \sum m(2,4,6,9,12) + D(1,3,13,14)$

(a) Find the simplest SOP expression of the function using a K-map (show your work).

		$x_1 \ x_2$			
		00	01	11	10
$x_3 \ x_4$	00	0	1	1	0
	01	$\emptyset_0$	0	$\emptyset_1$	1
	11	$\emptyset_0$	0	0	0
	10	1	1	$\emptyset_1$	0

Cost = 15 = 4 gates + 11 inputs

$$f = x_2 \bar{x}_4 + \bar{x}_1 x_3 \bar{x}_4 + x_1 \bar{x}_3 x_4$$

(b) Find the simplest POS expression of the function using a K-map (show your work).

		$x_1 \ x_2$			
		00	01	11	10
$x_3 \ x_4$	00	0	1	1	0
	01	$\emptyset_1$	0	$\emptyset_1$	1
	11	$\emptyset_1$	0	0	0
	10	1	1	$\emptyset_0$	0

Cost = 14 = 4 gates + 10 inputs

$$f = (\bar{x}_2 + \bar{x}_4)(\bar{x}_1 + \bar{x}_3)(x_2 + x_3 + x_4)$$

Question 5 (10%)

Consider the function $f = \sum m(1,3,4,9,11,13,15) + D(5,6,7,14)$

Find the simplest SOP and POS expressions of the function using K-maps (show your work).

SOP

$x_3 x_2$	$x_1 x_0$ 00	01	11	10
00	0	1	0	0
01	1	0	1	1
11	1	0	1	1
10	0	0	0	0

$$f = x_3 + \bar{x}_1 x_2$$

cost = 5 = 2 gates + 3 inputs

POS

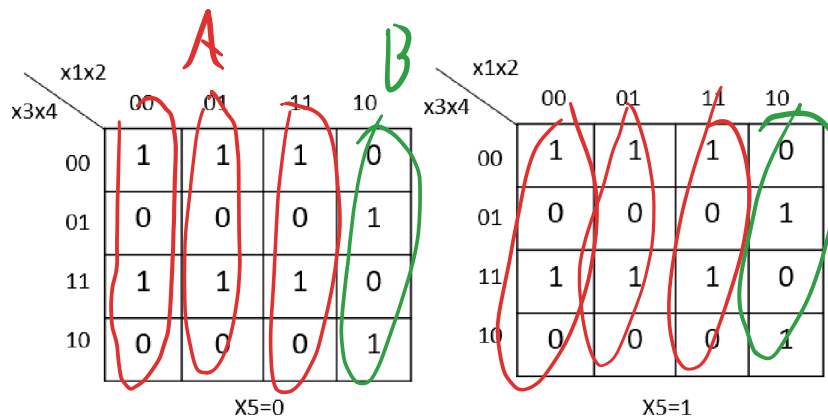
$x_3 x_2$	$x_1 x_0$ 00	01	11	10
00	0	1	0	0
01	1	0	1	1
11	1	0	1	1
10	0	0	0	0

$$f = (\bar{x}_3 + x_4)(\bar{x}_1 + x_4)(x_1 + x_4)$$

cost = 13 = 4 gates + 9 inputs

Question 6 (10%)

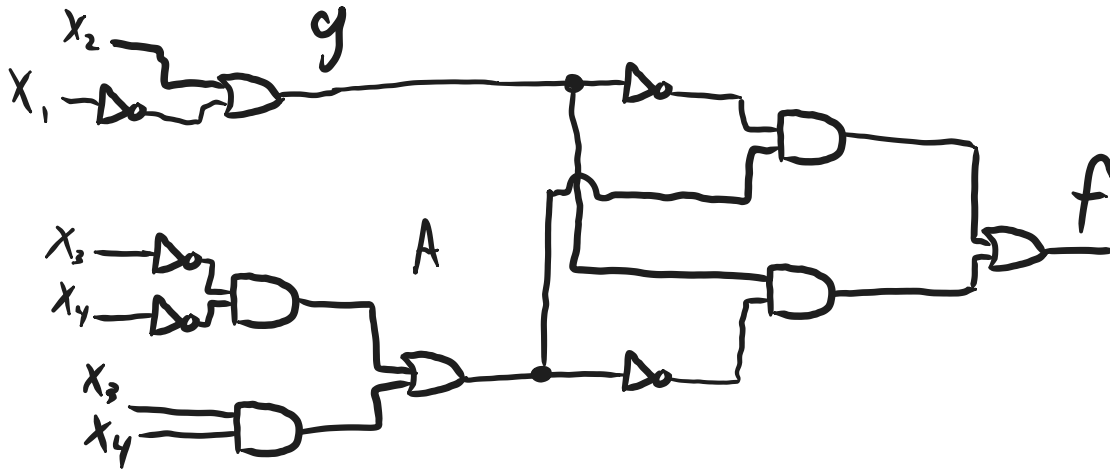
Given the K-map of a 5-variable function f below:



(a) Find the optimal (lowest cost) multi-level expression of f using functional decomposition.

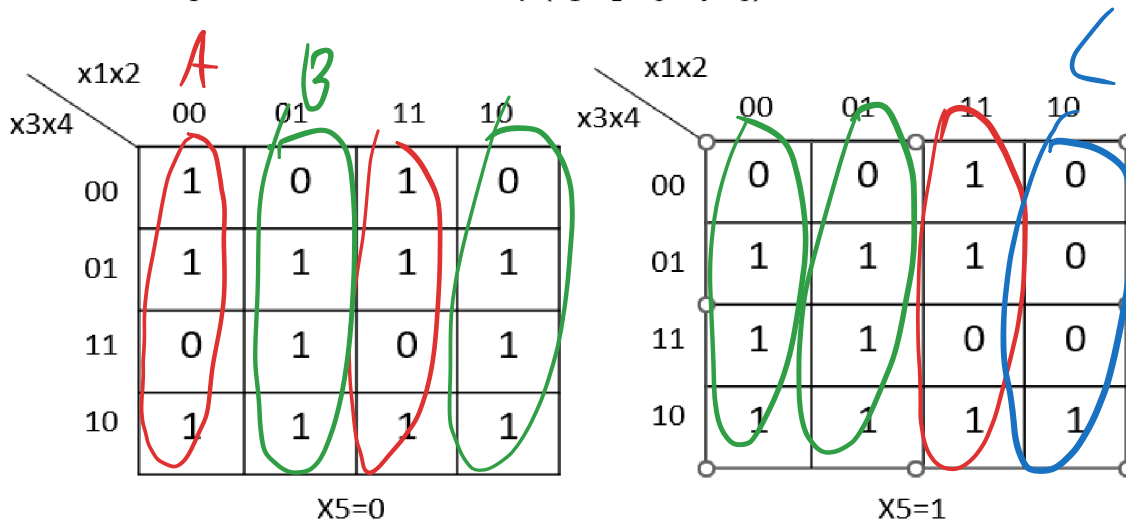
$$\begin{aligned}
 g &= (\bar{X}_1 + X_2) \rightarrow A & A &= (\bar{X}_3 \bar{X}_4 + X_3 X_4) \\
 g' &= X_1 \bar{X}_2 \rightarrow B & A' &= X_3 X_4 \text{ or } X_4 \\
 & & A &= X_3 X_4 \text{ or } X_4 \\
 & & & \\
 &= g A + g' A' \\
 f &= g(\bar{X}_1 \bar{X}_2 + X_1 X_2) + g'(X_1 \bar{X}_2 + \bar{X}_1 X_2) \\
 \text{cost}(g) &= 4 \text{ (Not assuming free complements)} & B = A' &= (\bar{X}_3 X_4 + X_4 \bar{X}_3) \\
 \text{cost}(A) &= 11 & & X \text{ or } \\
 \text{cost}(f) &= \text{cost}(g) + \text{cost}(A) + 11 = 26
 \end{aligned}$$

(b) Draw the circuit implementation from Q.1(a) with maximum fan-in equals to 2.



Question 7 (10%)

Given the K-map of a 5-variable function $f(x_1, x_2, x_3, x_4, x_5)$ below:



Using functional decomposition, find the expression of a multi-level implementation of f .
Hint: There could be more than two patterns in the K-map. Show your work and mark the patterns on the K-map.

$$g = \bar{x}_5 \bar{x}_1 \bar{x}_2 + x_1 x_2 \quad \rightarrow \quad A = (\bar{x}_3 \bar{x}_4) = \bar{x}_3 + \bar{x}_4$$

$$h = \bar{x}_1 x_2 + x_3 \bar{x}_1 \bar{x}_4 + \bar{x}_5 x_1 \bar{x}_2 \quad \rightarrow \quad B = x_3 + x_4$$

$$j = x_5 x_1 \bar{x}_2 \quad \rightarrow \quad C = x_3 \bar{x}_4$$

$$f = g(\bar{x}_3 + \bar{x}_4) + h(x_3 + x_4) + j(x_3 \bar{x}_4)$$

A
B
C

Question 8 (10%)

The VHDL code below describes the following system.

Inputs: 4-bit A and 2-bit D. Output: 1-bit Out

Functionality: When D = 00, Out = 0. When D = 01, Out = 1, if A is divisible by 2.

When D = 11, Out=1, if A is divisible by 3, When D = 10, Out=1, if A is divisible by 5. For all other cases, Out = 0.

Fill in the blanks of the provided code accordingly.

```

library IEEE;
use ieee.std_logic_1164.all;
use ieee.numeric_std.all;

entity vhdliawesome is
  port (
    A    : in std_logic_vector( 3  downto 0 );
    D    : in std_logic_vector( 1  downto 0 );
    Out  : out std_logic);
end vhdliawesome;

architecture description of vhdliawesome is
  signal div_by_2 : std_logic;
  signal div_by_3 : std_logic;
  signal div_by_5 : std_logic;
begin
  with A select div_by_2 <=
    '1' when "0000",
    '1' when "0010",
    '1' when "0100",
    '1' when "0110",
    '1' when "1000",
    '1' when "1010",
    '1' when "1100",
    '1' when "1110",
    '0' when OTHERS;

  with A select div_by_3 <=
    '1' when "0000",
    '1' when "0011",
    '1' when "0110",
    '1' when "1001",
    '1' when "1100",
    '1' when "1111",
    '0' when OTHERS;

  with A select div_by_5 <=
    '1' when "0000",
    '1' when "0101",
    '1' when "1010",
    '1' when "1111",
    '0' when OTHERS;

  with D select Out <=
    '0' when "00",
    div_by_2 when "01",
    div_by_3 when "11",
    div_by_5 when "10";

end description;

```